Honors Physics

Unit 2: Vectors and Projectile Motion

Student Workbook

Name: _____ Period: _____

How to use this workbook

- There is one section for each lesson. Each section lists the lesson's learning targets, the key ideas, and practice problems. Show all work in the space below each problem.
- Every learning target sits in one of four tiers: **Foundations**, **Core Skills**, **Target**, and **Extension**. A higher tier counts only when the tiers below it are secure, and you can reassess any tier after targeted practice.
- Problems marked **Check yourself** are a quick test of the section's main idea. If you cannot do one, revisit the key ideas before moving on.
- Use the target tracker at the back after every section and before every quiz.
- Course conventions: kinematic equations start from positions, such as $x_f = \frac{1}{2}at^2 + v_i t + x_i$ (we set $x_i = 0$; $y_i = 0$ for a launch from the ground, and $y_i = h$ with $y_f = 0$ for a launch from a height h). Displacement is $\Delta x = x_f - x_i$; when a question asks for it, the answer is written as Δx or Δy. Down is negative, and $a_y = -9.8\ \text{m/s}^2$. Ignore air resistance unless a problem says otherwise.

Every problem has a mark next to its number:

●	**Core**	Everyone does these. Together they cover the section's learning targets.
○	**More practice**	Extra reps. Do them when a core problem gave you trouble, or before a quiz.
★	**Challenge**	Harder, multi-step problems for when the core problems feel easy.

Contents

Section	Title	Tier
2.1	Scalars and Vectors	Foundations
2.2	Drawing and Adding Vectors	Foundations
2.3	Components: Breaking Vectors Apart	Foundations
2.4	Adding Vectors with Components	Core Skills
2.5	Reference Frames and Relative Velocity	Core Skills
2.6	Relative Velocity with Angles	Target
	Get Ready for Quiz 1	Quiz 1: Vectors and Relative Velocity
2.7	Independence of Motion: Hit-the-Target Lab	Foundations
2.8	Horizontal Launches: x and y Columns	Core Skills
2.9	Velocity During Flight (Marble Off the Table)	Core Skills
	Get Ready for Quiz 2	Quiz 2: Horizontal Launches
2.10	Angled Launches on Level Ground	Target
2.11	Cliffs, Ledges, and Velocity at Impact	Target
2.12	Working Backward and Deriving in Symbols	Target
	Get Ready for Quiz 3	Quiz 3: Angled Launches
	Putting It All Together	Target
	Review and Reassessment Workshop (separate handout)	All tiers
	Unit test (no workbook page)	Foundations, Core Skills, Target
	Projectile Launch Project: Planning Pages (separate handout)	Target / Extension
	Reference sheet, vocabulary, target tracker, extension menu	

Reference sheet

Vectors

Vector components	Resultant vectors
$A_x = A\cos\theta$ $A_y = A\sin\theta$ θ is measured from the +x axis. Left and down components are negative.	Add components: $R_x = A_x + B_x, R_y = A_y + B_y$ Magnitude: $R = \sqrt{R_x^2 + R_y^2}$ Direction: $\theta = \tan^{-1}\left(\frac{\lvert R_y\rvert}{\lvert R_x\rvert}\right)$

SOH-CAH-TOA: $\sin\theta = \dfrac{\text{opposite}}{\text{hypotenuse}}, \cos\theta = \dfrac{\text{adjacent}}{\text{hypotenuse}}, \tan\theta = \dfrac{\text{opposite}}{\text{adjacent}}$

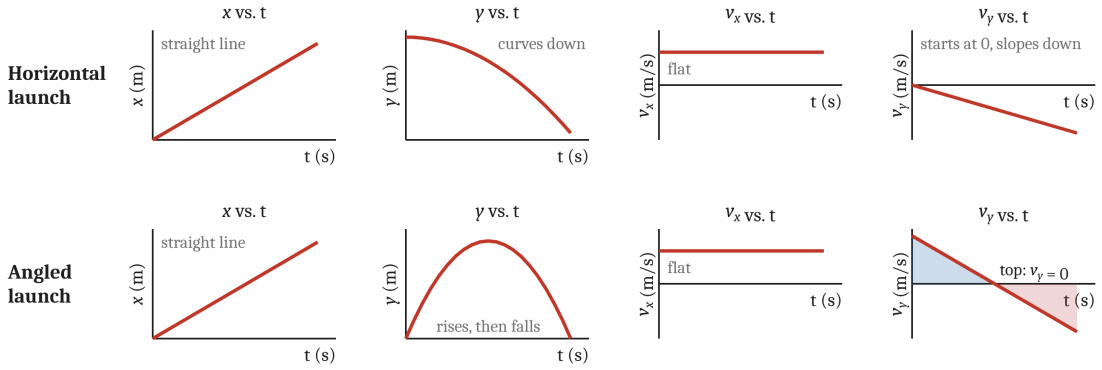
Pythagorean theorem: $c^2 = a^2 + b^2$

Relative velocity

- Add velocities as vectors: $v_{\text{object rel. ground}} = v_{\text{object rel. medium}} + v_{\text{medium rel. ground}}$
- **Assume:** each velocity is constant, and the medium (water or air) carries the object along with it.
- **Perpendicular velocities:** find the resultant with the Pythagorean theorem and its direction with $\tan^{-1}$.
- **Crossing time:** only the velocity component straight across the river sets it.
- **Drift:** the current carries the boat downstream while it crosses.
- **Landing directly across:** aim upstream so the upstream part of the boat's velocity cancels the current.

Motion graphs

Top row: a horizontal launch. Bottom row: an angled launch on level ground.



- **Slope of a position graph** (x–t or y–t) = velocity (v_x or v_y).
  - A straight position graph means constant velocity; a curved one means the velocity is changing.
- **Slope of a velocity graph** (v–t) = acceleration. v_x–t is flat ($a_x = 0$); v_y–t slopes at –9.8 m/s².
- **Area under a velocity graph** (between the line and the time axis) = change in position, Δx or Δy.
  - Area above the axis counts as positive; area below counts as negative.

Projectile motion: x world and y world

Ignore air resistance. Down is negative. Time t is the same in both columns and links them.

x world (horizontal)	**y world (vertical)**
$v_{ix} =$	$v_{iy} =$
$v_{fx} =$	$v_{fy} =$
$a_x = 0$	$a_y = -9.8\ \text{m/s}^2$
$x_i =$	$y_i =$
$x_f =$	$y_f =$
t =	t =
Every projectile	
Assume: $a_x = 0$, so $v_{fx} = v_{ix}$ the whole flight. **Equation:** $x_f = v_{ix}t + x_i$	**Assume:** $a_y = -9.8\ \text{m/s}^2$. At the top, $v_y = 0$. **Equations:** $y_f = \frac{1}{2}a_y t^2 + v_{iy}t + y_i$ $v_{fy} = a_y t + v_{iy}$ $v_{fy}^2 = v_{iy}^2 + 2a_y(y_f - y_i)$
Horizontal launch	
Assume: $v_{ix} =$ launch speed, $x_i = 0$.	**Assume:** $v_{iy} = 0$, $y_i = h$ (launch height), $y_f = 0$ (the ground).
Angled launch from the ground	
Assume: $v_{ix} = v_i \cos\theta$, $x_i = 0$. Range uses the total time.	**Assume:** $v_{iy} = v_i \sin\theta$, $y_i = 0$, $y_f = 0$. It lands with $v_{fy} = -v_{iy}$.
Angled launch from a cliff	
Assume: $v_{ix} = v_i \cos\theta$, $x_i = 0$.	**Assume:** $v_{iy} = v_i \sin\theta$, $y_i = h$ (cliff height), $y_f = 0$ (the ground). v_{fy} is negative at landing.

Solving routine

1. Sketch the path. Label the launch point, the landing point, and the ground ($y = 0$).
2. Fill in both columns. Mark what you know and what you want.
3. Find t from the column that has enough knowns (usually the y world).
4. Use that t in the other column to finish.

Common mistakes

- $v_{iy} = 0$ only for a horizontal launch.
- a_x is always 0. Only the y world speeds up or slows down.
- Cosine goes with the side next to the angle: $v_{ix} = v_i \cos\theta$ when θ is measured from the horizontal.
- At the top, $v_y = 0$ but v_x is not, so the ball is still moving.
- v_f is not 0 at the ground. The equations describe the ball just before it lands, while it is still falling fast; the stop happens during the impact, which the equations do not model. Use $y_f = 0$, not $v_{fy} = 0$.

Velocity at any moment

$v_x = v_{ix}$ the whole flight and $v_y = a_y t + v_{iy}$, so $v = \sqrt{v_x^2 + v_y^2}$ and $\theta = \tan^{-1}\left(\dfrac{|v_y|}{v_x}\right)$ (below the horizontal when v_y is negative, above it when positive).

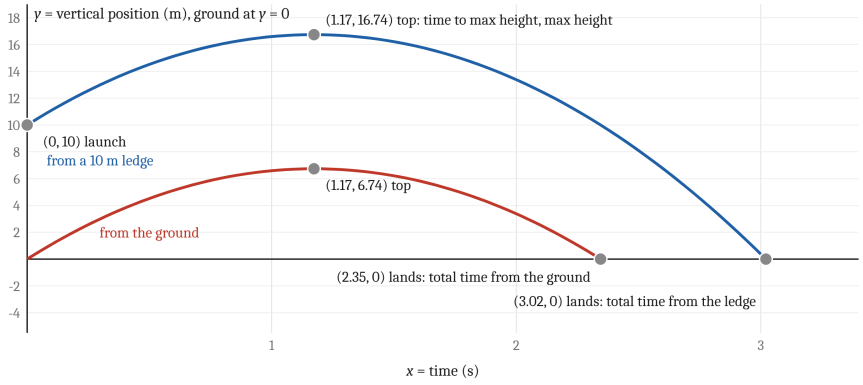
Solving when the y equation is a quadratic

When you know the heights but not the time, the y-world equation $y_f = \frac{1}{2}a_y t^2 + v_{iy}t + y_i$ is a quadratic in t. Move everything to one side: $0 = \frac{1}{2}a_y t^2 + v_{iy}t + (y_i - y_f)$. Solve it either way below; both give the same time. Example: a ball launched at 15 m/s at 50° ($v_{iy} = 11.49$ m/s), once from the ground and once from the top of a 10 m ledge.

Method 1: graph it in Desmos

Desmos (desmos.com/calculator) graphs the y-world equation so you can click to read off key points. In Desmos, x stands for time t and y stands for vertical position.

1. Set Desmos to degrees: click the wrench (top right) and choose **Degrees**.
2. Type the y-world equation with your numbers, using x for time (the pattern is $y_f = \frac{1}{2}a_y t^2 + v_{iy}t + y_i$). From the ground: `y=0.5(-9.8)x^2+15sin(50)x+0`. From a height, the last number is the launch height: `y=0.5(-9.8)x^2+15sin(50)x+10`.
3. Zoom out until you see the whole arc. Click the curve: gray points appear. Click a point to see its coordinates (time, height).
4. With the ground at $y = 0$, the ball lands where the curve crosses the x axis. To land on a ledge instead, add the ledge height as a second line, such as `y=6`, and click where it crosses the curve.
5. For the horizontal distance, type v_{ix} times the time you found: `15cos(50)*2.345` gives 22.6 m from the ground, and `15cos(50)*3.021` gives 29.1 m from the ledge.



To find	Click on	Example reading
Time to max height and max height	the top of the curve	Ground: (1.17, 6.74). Ledge: (1.17, 16.74), so 16.74 m above the ground.
Total time, launched from the ground	where the curve crosses the x axis again	(2.35, 0): 2.35 s
Total time, launched from a height	where the curve crosses the x axis (the ground)	(3.02, 0): 3.02 s
Range or horizontal final position	type v_{ix} × time on a new line	Ground: 22.6 m. Ledge: 29.1 m.

- Horizontal launch: $v_{iy} = 0$, so type `y=0.5(-9.8)x^2+h` with the launch height for h, and click where the curve crosses `y=0`.
- Time to the top another way: graph v_y with `y=-9.8x+15sin(50)`. It crosses the x axis at the time of max height.

Method 2: the quadratic formula

$$t = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \text{ with } a = \tfrac{1}{2}a_y = -4.9\ \text{m/s}^2, b = v_{iy}, c = y_i - y_f$$

- From the ledge: $a = -4.9, b = 11.49, c = 10 - 0 = 10$, so $t = 3.02$ s or $t = -0.68$ s. Keep the positive root, 3.02 s; a negative time is before the launch.
- If both roots are positive (landing on a ledge above the launch point), the ball passes that height twice. The later root is the landing on the way down.
- From the ground, $c = 0$, so the roots are $t = 0$ (the launch) and $t = 2.35$ s (the landing).

Reasoning and lab skills

Paragraph argument

- **Claim:** answer the question in one sentence.
- **Evidence:** a physics principle or equation (for example $a_x = 0$, or independence of motion).
- **Reasoning:** connect the evidence to this situation.
- Write 3–5 complete sentences.

Checking your answer

- **Units:** do they match what the question asks for?
- **Signs:** down is negative.
- **Limiting case:** try $\theta \to 0°$, $\theta \to 90°$, or a very large speed. Does the result do what physics predicts?
- **Size:** is the number reasonable for the situation?

Designing an experiment

- List each quantity you measure and the tool you use for it.
- **Independent variable:** what you change.
- **Dependent variable:** what you measure.
- **Controlled variables:** what you keep the same.
- Run repeated trials and average them.
- Choose a graph that makes a straight line, and say what its slope (or intercept) tells you.

Lab results

- **Percent difference:** $\dfrac{|\text{measured} - \text{predicted}|}{\text{predicted}} \times 100\%$
- Name a source of error, and say which way it pushes the result (lands short or long, too big or too small).

Vocabulary

Term	Definition
Scalar	A quantity with magnitude only (distance, speed, time)
Vector	A quantity with magnitude and direction (change in position, velocity, acceleration)
Magnitude	The size of a vector, always positive
Direction	The way a vector points, given as an angle from a reference axis
Resultant	The single vector equal to the sum of two or more vectors
Component	The horizontal (x) or vertical (y) part of a vector
Angle of launch	The angle of the initial velocity above or below the horizontal
Projectile	An object moving through the air under gravity alone
Trajectory	The curved path a projectile follows
Range	The horizontal change in position from launch to landing
Independence of motion	Horizontal and vertical motions happen at the same time but do not affect each other
Relative velocity	An object's velocity as measured from a particular reference frame, such as the riverbank or the moving water

Section 2.1 · Scalars and Vectors

Tier: Foundations **In class:** Vocabulary sort; scalar and vector practice

Learning targets

- Key terms I can tell a scalar from a vector and use magnitude, direction, and resultant correctly.

Key ideas

- A **scalar** has magnitude only (distance, speed, time, mass).
- A **vector** has magnitude and direction (change in position, velocity, acceleration, force).
- Change in position (displacement) starts from positions: $\Delta x = x_f - x_i$. After that, just write Δx. It can be negative; distance traveled cannot.

1. ● Label each quantity S (scalar) or V (vector).

a) 25 m/s ___ b) 25 m/s north ___ c) 3.0 s ___ d) 9.8 m/s^2 down ___

e) 12 km ___ f) 12 km southwest ___ g) mass, 4.0 kg ___ h) temperature, 21 °C ___

2. ● Mrs. Gordon runs exactly one lap of a 400 m track and finishes at the starting line. What distance did Mrs. Gordon travel? What is the change in position, Δx? Explain the difference.

3. ● A car drives 60 km east, then 20 km west. The whole trip takes 1.0 h. Take east as positive.

a) Distance traveled b) Change in position c) Average speed d) Average velocity

4. ○ Two vectors have magnitudes 6 m and 8 m. What is the largest their resultant could be? The smallest? Sketch how each one happens.

5. ● **Check yourself.** Write one sentence explaining why speed is a scalar but velocity is a vector.

Section 2.2 · Drawing and Adding Vectors

Tier: Foundations **In class:** Vector walk / campus scavenger hunt

Learning targets

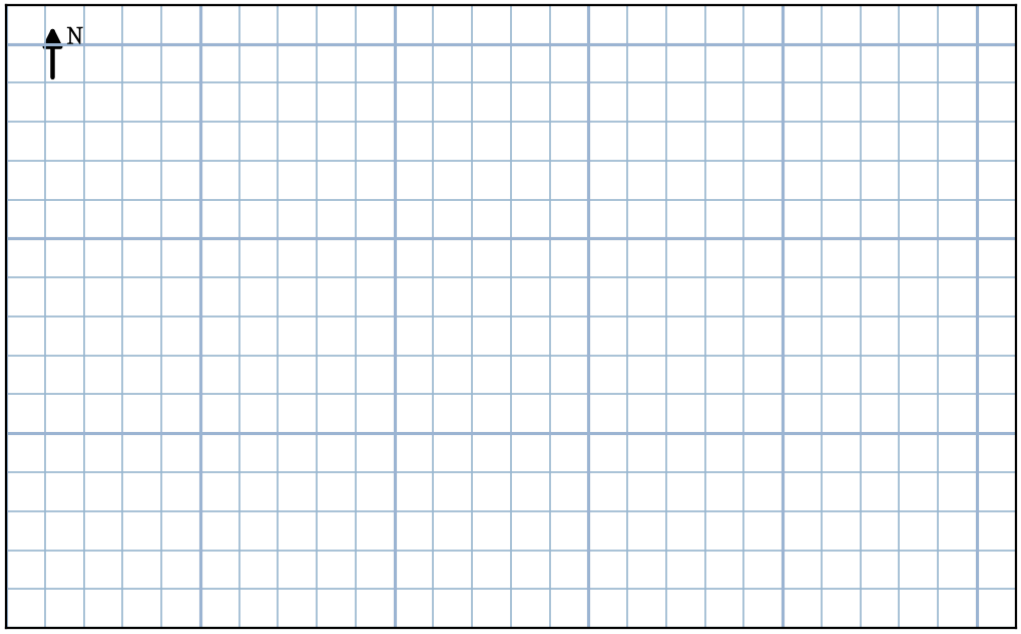
- F1 I can draw vectors to scale and add them tip-to-tail.
- F3 I can use the Pythagorean theorem and inverse tangent to find a resultant's magnitude and direction.
- F6 I can find the resultant of two perpendicular vectors.

Key ideas

- Draw each vector to scale with a ruler and protractor. Put the tail of the next vector at the tip of the last.
- The **resultant** goes from the tail of the first vector to the tip of the last.
- For perpendicular vectors: $R = \sqrt{A^2 + B^2}$ and $\theta = \tan^{-1}\left(\dfrac{\text{opposite}}{\text{adjacent}}\right)$.
- Give direction as an angle from a reference direction, such as 37° N of E (start facing east, turn 37° toward north).

1. ● Scale drawing. Use the grid (each square is 5 m). A student walks 40 m east, then 30 m north. Draw both vectors tip-to-tail, draw the resultant, and measure its length and angle.

Measured: R = _____ at _____ Calculated: R = _____ at _____



2. ● Mr. Jantz walks 3 blocks east and then 4 blocks north to get coffee. Find the magnitude and direction of the resultant.

3. ○ A hiker walks 12 km north and then 5 km west. Find the resultant.

4. ● **Vector walk preview.** On the walk, a group goes 25 m south and then 60 m east. Find the resultant, then describe the single straight walk that would bring them back to the start.

5. ○ You walk 120 m east, then turn and walk north until you are 200 m from your starting point. How far north did you walk? In what direction would you head to walk straight home?

6. ○ Does the order you add vectors in change the resultant? Test it by redrawing problem 2 with north first.

7. ○ Mr. Peck throws a ball for Phoebe. Phoebe runs 8.0 m west and then 6.0 m south to get it. Find the magnitude and direction of her resultant.

8. ★ **Challenge.** You walk 5.0 m north and then some distance east. You end up 13 m from where you started. How far east did you walk?

9. ● **Check yourself.** Vectors of 5 m east and 3 m west add to what resultant?

Section 2.3 · Components: Breaking Vectors Apart

Tier: Foundations **In class:** Component-resolving practice with protractors

Learning targets

- F2 I can use SOH-CAH-TOA to resolve a vector into x and y components.
- F3 I can find a vector's magnitude and direction from its components.

Key ideas

- For a vector of magnitude A at angle θ above the +x axis: $A_x = A\cos\theta$ and $A_y = A\sin\theta$.
- Signs come from the picture: left and down are negative.
- Going back: $A = \sqrt{A_x^2 + A_y^2}$ and $\theta = \tan^{-1}\left(\dfrac{|A_y|}{|A_x|}\right)$. Then name the quadrant in words.
- Always sketch the triangle first. The side next to the angle uses cosine.

1. ● Resolve each vector into x and y components. Sketch each one.

a) 20 m/s at 30° above the horizontal

b) 50 m/s at 37° above the horizontal

c) 15 m at 60° south of east

d) 120 km at 25° west of north

2. ● Mr. White kicks a soccer ball at 25 m/s at 53° above the ground. Find the horizontal and vertical components of its initial velocity, v_{ix} and v_{iy}.

3. ○ A skier moves at 12 m/s straight down a 25° slope. How fast is the skier moving horizontally? How fast is the skier losing height? How long does it take to drop 50 m in height?

4. ● A vector has components $x = -12$ m and $y = +5$ m. Find its magnitude and direction.

5. ○ As the launch angle of a 20 m/s ball increases from 0° to 90°, what happens to v_{ix}? To v_{iy}? At what angle are they equal?

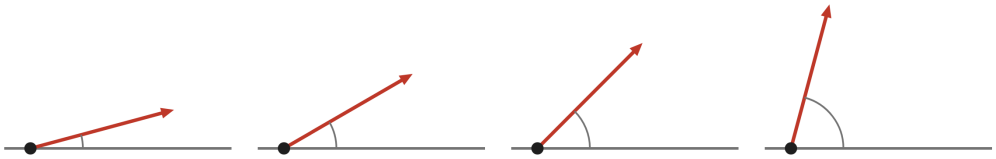
6. ○ A displacement of 18 m points 30° west of south. Find its x and y components (east and north are positive).

7. ○ A ramp launches a marble at 3.0 m/s, 20° above the horizontal. Find v_{ix} and v_{iy}.

8. ★ **Challenge.** Mrs. Kraus jumps off the ground at 40° above the horizontal with a vertical velocity component of 3.0 m/s. Find her launch speed and her horizontal velocity component.

9. ● **Symbolic.** A vector of magnitude A points at angle θ above the +x axis. Write A_x and A_y in terms of A and θ. Then show that $A_x^2 + A_y^2 = A^2$. Which right-triangle fact did you just use?

10. ● **Ranking.** Mrs. Luksic hits four softballs, each at 20 m/s, at the angles shown. Rank the balls, greatest first.



Ball A: 20 m/s at 15° Ball B: 20 m/s at 30° Ball C: 20 m/s at 45° Ball D: 20 m/s at 75°

a) v_{ix}: Greatest _____ _____ _____ _____ Least ☐ All equal

b) v_{iy}: Greatest _____ _____ _____ _____ Least ☐ All equal

c) Which two balls have components that swap places? _____ and _____ Explain why.

11. ● **Check yourself.** A student writes $v_{ix} = 30 \sin 40^\circ$. When is that correct?

Section 2.4 · Adding Vectors with Components

Tier: Core Skills **In class:** Component-addition problem set

Learning targets

- C1 I can add non-perpendicular vectors using components.

Key ideas

- Resolve every vector into x and y components and record them in a table.
- Add the x column to get R_x and the y column to get R_y.
- Then $R = \sqrt{R_x^2 + R_y^2}$ and $\theta = \tan^{-1}\left(\dfrac{|R_y|}{|R_x|}\right)$. Sketch R_x and R_y to name the direction.

1. ● $A = 30$ m east and $B = 20$ m at 60° N of E. Fill in the table and find $A + B$.

Vector	x component	y component
A		
B		
R		

2. ● A hiker walks 4.0 km at 30° N of E, then 6.0 km at 45° W of N. Find the hiker's resultant change in position.

3. ○ What single walk would take the hiker in problem 2 straight back to the start?

4. ○ Add three vectors: 10 m north, 15 m at 40° S of E, and 8.0 m west.

5. ○ Without a calculator, add $A = 5.0$ m east, $B = 5.0$ m north, and $C = 5.0$ m west.

6. ○ A rescue boat travels 3.0 km at 30° N of E, then 5.0 km at 20° W of N. A helicopter at the boat's starting point flies straight to the boat at 120 km/h. Find the helicopter's heading and the flight time in minutes.

7. ★ **Challenge.** $A + B = 10$ m north, and $A = 6.0$ m at 30° N of E. Find B (magnitude and direction).

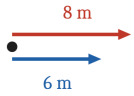
8. ● **Symbolic.** Dr. Shafer and a partner drag a heavy sled with two ropes. Each pulls with the same force F (a vector, in newtons), one rope at angle θ to the left of straight ahead and one at θ to the right.

a) Find the forward and sideways components of each pull. b) Show that the combined pull is $2F\cos\theta$ straight ahead.

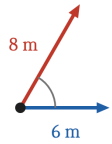
c) What is the combined pull when $\theta = 0°$? When $\theta = 90°$? Why should the two pullers keep their ropes close together?

9. ● **Ranking.** In each case a 6 m vector and an 8 m vector are added. Only the angle between them changes. Rank the cases by the magnitude of the resultant, greatest first, without calculating.

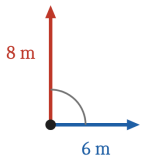
Case A: angle 0°



Case B: angle 60°



Case C: angle 90°



Case D: angle 180°



Resultant: Greatest _____ _____ _____ _____ Least ☐ All equal

Check case C with the Pythagorean theorem: R = _____ Explain the pattern.

10. ● **Check yourself.** Could two vectors of 5 m and 12 m add to 20 m? Explain.

Section 2.5 · Reference Frames and Relative Velocity

Tier: Core Skills **In class:** Boat-and-river vector problems

Learning targets

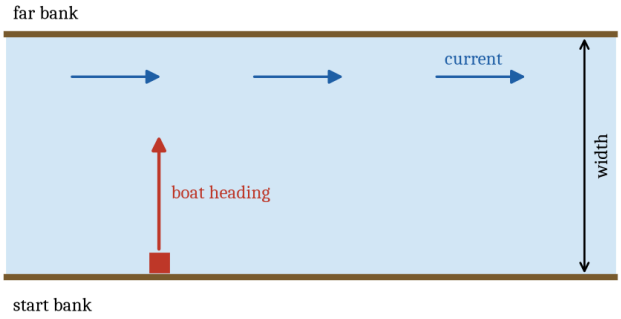
- C5 I can solve relative-velocity problems with perpendicular vectors (a boat crossing a river).

Key ideas

- Velocity depends on the reference frame: the boat's velocity relative to the water is not its velocity relative to the bank.
- $v_{\text{boat rel. ground}} = v_{\text{boat rel. water}} + v_{\text{water rel. ground}}$
- Crossing a river: $t = \dfrac{\text{width}}{v_{\text{across}}}$, where v_{across} is the velocity component straight across. The current does not change that time.
- Downstream drift: $\text{drift} = v_{\text{current}}\, t$

1. ● At the airport, Mrs. Gordon jogs at 1.5 m/s relative to a moving walkway that moves at 1.0 m/s. What is her speed relative to the ground if she jogs a) with the walkway? b) against it?

2. ● A boat heads straight across a 120 m wide river at 4.0 m/s relative to the water. The current flows at 1.5 m/s.
a) Find the boat's velocity relative to the riverbank (magnitude and direction).
b) How long does it take to cross? c) How far downstream does it land?



3. ○ A swimmer who swims 1.5 m/s in still water points straight across a 90 m river and lands 48 m downstream. Find the speed of the current. The swimmer then walks back along the bank at 1.2 m/s to the spot directly across from the start. Find the total trip time.

4. ○ A plane points due north with an airspeed of 220 m/s. A 40 m/s wind blows toward the east. Find the plane's velocity relative to the ground.

5. ○ Rain falls straight down at 8.0 m/s. You drive at 15 m/s. Relative to you, how fast is the rain moving, and at what angle from vertical do its streaks appear on the side window?

6. ● **Symbolic.** A boat with speed v_b relative to the water points straight across a river of width w. The current is v_c. Write the crossing time, the downstream drift, and the angle of the path from straight across. Which of the three change if the current doubles?

7. ● **Qualitative.** Does a stronger current help, hurt, or not affect a swimmer who points straight across? Answer separately for the crossing time and the landing point, and explain.

8. ● **Check yourself.** To cross a river in the least time, which way should you point the boat?

Section 2.6 · Relative Velocity with Angles

Tier: Target **In class:** River-crossing and crosswind stations

Learning targets

- T5 I can solve relative-velocity problems with angles (aiming upstream; a plane in a crosswind).

Key ideas

- To land directly across, aim upstream so the upstream part of the boat's velocity cancels the current: $v_{\text{boat}} \sin\theta = v_{\text{current}}$.
- Then the speed across is $v_{\text{boat}} \cos\theta$, and time = width ÷ speed across.
- For a plane, the heading is where it points. Ground velocity: $v_{\text{plane rel. ground}} = v_{\text{plane rel. air}} + v_{\text{air rel. ground}}$

1. ● A boat moves at 5.0 m/s relative to the water. The current is 3.0 m/s, and the river is 80 m wide.

a) At what angle upstream must it aim to land directly across?

b) What is its speed relative to the bank? c) How long does the crossing take?

2. ● A boat (4.0 m/s relative to the water) aims 20° upstream on a 60 m river with a 2.0 m/s current. How long does it take to cross, and where does it land?

3. ● A pilot wants to travel due north. The plane's airspeed is 250 m/s and the wind blows toward the east at 50 m/s. Find the heading and the ground speed. How long does a 500 km trip take?

4. ○ A plane heads 30° N of E at 180 m/s through a wind blowing toward the south at 40 m/s. Find its velocity relative to the ground.

5. ● A swimmer who swims at 1.0 m/s tries to cross a river whose current is 1.5 m/s. Can she land directly across? Explain using vectors.

6. ○ A boat moving at 3.0 m/s relative to the water aims 30° upstream in a 1.5 m/s current. Does it land directly across? What is its speed across the river?

7. ● **Symbolic.** A boat with speed v_b relative to the water must land directly across a river of width w with current $v_c (v_c < v_b)$. Show that the aim angle upstream is $\theta = \sin^{-1}\left(\dfrac{v_c}{v_b}\right)$ and the crossing time is $t = \dfrac{w}{\sqrt{v_b^2 - v_c^2}}$. What happens to t as v_c gets close to v_b? Why does that make sense?

8. ○ **Ranking.** Four planes, each with an airspeed of 200 m/s, must travel due north. Each flies in a different wind that blows toward the east. Rank the planes, greatest first.

Plane	Airspeed	Wind (toward the east)
A	200 m/s	0 m/s
B	200 m/s	20 m/s
C	200 m/s	40 m/s
D	200 m/s	60 m/s

a) Angle the plane must turn west of north: Greatest _____ _____ _____ _____ Least ☐ All equal

b) Ground speed: Greatest _____ _____ _____ _____ Least ☐ All equal

Explain both rankings.

9. ● **Check yourself.** Rain streaks on a car's side window make a 60° angle with the vertical when the car moves at 20 m/s. How fast is the rain falling?

Get Ready for Quiz 1

Tier: Quiz 1: Vectors and Relative Velocity **In class:** Quiz 1: Vectors and relative velocity (full period)

About this page

- Quiz 1 covers F1, F2, F3, F6, C1, C5, and T5. Complete these pages before the quiz, and rate yourself in the tracker first.

1. ● Define resultant and component in your own words.

2. ● Resolve 35 m/s at 25° below the horizontal (pointing right) into components.

3. ● Add 8.0 m at 0° and 5.0 m at 120° (measured counterclockwise from +x).

4. ● A canoe heads straight across a 45 m river at 3.0 m/s; the current is 1.2 m/s. Find the time to cross, the drift, and the canoe's velocity relative to the bank.

5. ● **Ranking.** Four planes each head due north at an airspeed of 150 m/s. The wind is 30 m/s in each case, from a different direction. Rank the planes by ground speed, greatest first.

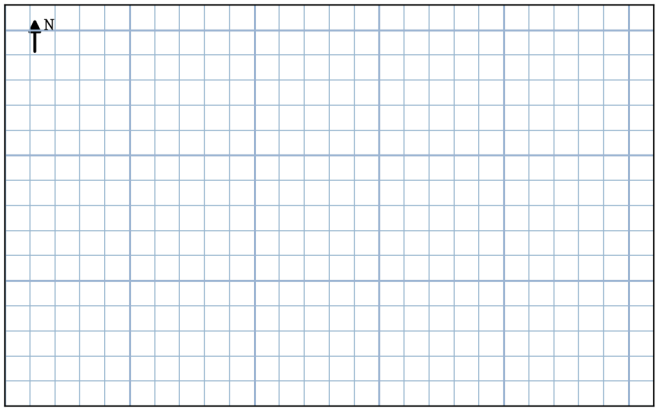
Plane	Heading	Wind blows toward
A	North	East
B	North	West
C	North	South
D	North	North

Ground speed: Greatest _____ _____ _____ _____ Least ☐ All equal

Explain why A and B compare the way they do.

6. ● A boat (6.0 m/s relative to the water) must land directly across a 90 m river with a 2.5 m/s current. Find its aiming angle and crossing time.

7. ○ Phoebe chases a ball 12 m at 30° N of E, then runs 8.0 m due south. On the grid (1 square = 1 m), draw both moves tip-to-tail and measure Phoebe's resultant displacement. Then check it with components.



8. ○ Can a 5.0 m vector and an 8.0 m vector add to a resultant of a) 2.0 m, b) 13 m, c) 10 m? Explain each with a sketch.

9. ○ **Symbolic.** A boat with speed v_b relative to the water points straight across a river of width w with current v_c. Write expressions for the crossing time and the downstream drift. What happens to each if the current doubles?

10. ★ **Challenge.** A pilot must fly to a city 400 km away at 30° N of E. The plane's airspeed is 220 km/h, and a 40 km/h wind blows toward the south. Find the heading the pilot must use and the time the trip takes.

Section 2.7 · Independence of Motion: Hit-the-Target Lab

Tier: Foundations **In class:** Hit-the-Target Lab: predict where the ball lands, then test

Learning targets

- F4 I can explain that the horizontal and vertical motions of a projectile are independent.
- F5 I can state that, ignoring air resistance, horizontal velocity is constant and $a_y = -9.8\ \text{m/s}^2$.

Key ideas

- Today's lab: measure your launch, predict exactly where the ball will land, put a cup there, and test it.
- Horizontal: $a_x = 0$, so v_x never changes. Vertical: $a_y = -9.8\ \text{m/s}^2$, just like a dropped object.
- The only thing the two directions share is **time**. The y world tells you how long the ball falls; the x world tells you how far it goes in that time.

1. ● Hit-the-Target Lab. Goal: predict where a ball rolling off your table will land, place a cup there, and land the ball in the cup.

Materials: ramp, marble or steel ball, meter stick, masking tape, paper cup, phone with slow-motion video, plumb line (or a second ball to drop).

Procedure:

a) Tape the ramp to the table so the ball leaves the edge moving horizontally. Mark one release spot on the ramp and use it every time.

b) Measure the table height h (floor to the bottom of the ball at the edge) and the cup height.

h = _____ cup height = _____

c) Put two tape marks 0.20 m apart on the tabletop just before the edge. Film three rolls in slow motion and count the frames between the marks. Time = frames ÷ frame rate; $v_{ix} = 0.20\ \text{m} \div \text{time}$.

Trial	Frames between marks	Frame rate (fps)	Time (s)	v_{ix} (m/s)
1				
2				
3				
Average				

d) **Predict.** Set up x and y columns. The ball only has to fall to the cup's rim, so use $y_i = h$ and $y_f =$ cup height. Find the fall time and the landing distance x_f.

t = _____ $x_f =$ _____

e) Use the plumb line to mark the floor directly below the table edge. Measure x_f from that mark and place the cup's center there. Have your teacher check the cup before you launch.

f) **Test.** Launch three times from the release spot. Record where each ball lands (or "in the cup").

Trial 1: _____ Trial 2: _____ Trial 3: _____ Average: _____

g) Find the percent difference between your average landing and your prediction.

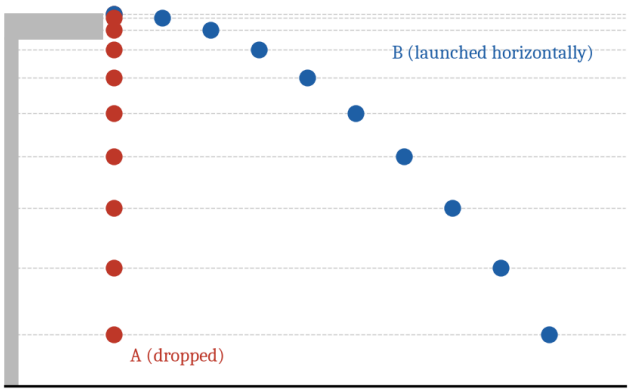
h) Which measurement (h, the cup height, or v_{ix}) has the biggest effect on your prediction? Name one change that would make the next prediction better.

i) **Source of error.** Name one source of error other than a misread measurement. Would it make the ball land short of the cup or past it? Explain why.

2. ● Concept check. The picture shows ball A dropped and ball B launched horizontally at the same instant. Dots are positions every 0.10 s.

a) Compare the heights of A and B at each instant. b) What does B's equal horizontal spacing tell you about v_x?

c) The vertical gaps grow each interval. What does that tell you about v_y?



3. ● Concept check. Ball A rolls off a table at 3.0 m/s. Ball B rolls off the same table at 6.0 m/s. Without calculating, compare a) their fall times, b) their landing distances, and c) their v_y just before landing. Explain each with the x world or the y world.

4. ● A ball leaves a 12 m ledge at 5.0 m/s horizontally. Fill in the table. The launch point is $x_i = 0$, $y_i = 12$ m, the ground is $y = 0$, and down is negative.

t (s)	v_x (m/s)	v_y (m/s)	x (m)	y (m)
0				
0.50				
1.0				
1.5				

5. ○ A plane flying level at constant speed drops a supply crate. Ignoring air resistance, where is the crate relative to the plane when it hits the ground? Explain.

6. ★ **Challenge.** In the lab, a group forgets the cup's height. Their table is 0.90 m high, the cup is 0.12 m tall, and $v_{ix} = 2.4$ m/s. Where do they place the cup, and where should it go? Does the ball fall short or go long?

7. ● **Symbolic.** A ball rolls off a table of height h at speed v. Show that it lands a distance $x = v\sqrt{\dfrac{2h}{g}}$ from the table. By what factor does x change if h is made four times larger? If v is doubled?

8. ○ **Ranking.** Four balls roll off tables. Rank them, greatest first, and explain each ranking with the x world or the y world.

Ball	Table height h	Launch speed v
A	1.0 m	2.0 m/s
B	1.0 m	4.0 m/s
C	4.0 m	1.0 m/s
D	0.25 m	4.0 m/s

a) Fall time: Greatest _____ _____ _____ _____ Least ☐ All equal

b) Landing distance: Greatest _____ _____ _____ _____ Least ☐ All equal

Explain.

9. ○ Ivy knocks a toy off a table. It leaves the edge moving horizontally at 2.0 m/s and lands 0.45 s later. How far from the table does it land?

10. ○ A marble rolls off a 0.80 m table at 1.5 m/s. How far from the table edge should a target on the floor go?

11. ○ Which of these changes the fall time in the lab: doubling v_{ix}, using a heavier ball, or raising the table? Explain.

12. ● **Check yourself.** Two marbles leave the same table at the same moment, one at 1.0 m/s and one at 3.0 m/s. Which lands first? Which lands farther away?

Section 2.8 · Horizontal Launches: x and y Columns

Tier: Core Skills **In class:** Notes and worked examples; TIPERs: two rocks

Learning targets

- C2 I can organize a projectile problem into x and y columns linked by the shared time.
- C3 I can find the time to land and the horizontal final position of a horizontally launched projectile.

Key ideas

- Set up two columns. x: v_{ix}, $a_x = 0$, x_i, x_f, t. y: $v_{iy} = 0$, $a_y = -9.8\ \text{m/s}^2$, $y_i = h$ (the launch height), $y_f = 0$ (the ground), t.
- The column with three knowns solves for t. Then t carries over to the other column.

1. ● A ball rolls off a 0.90 m high table at 2.5 m/s. Use the columns to find the time to land and how far from the table it lands.

x (horizontal)	y (vertical)
$v_{ix} =$	$v_{iy} =$
$v_{fx} =$	$v_{fy} =$
$a_x =$	$a_y =$
$x_i =$	$y_i =$
$x_f =$	$y_f =$
t =	t =

2. ● A stone is thrown horizontally at 15 m/s from a 45 m cliff. Find the time in the air and where it lands.

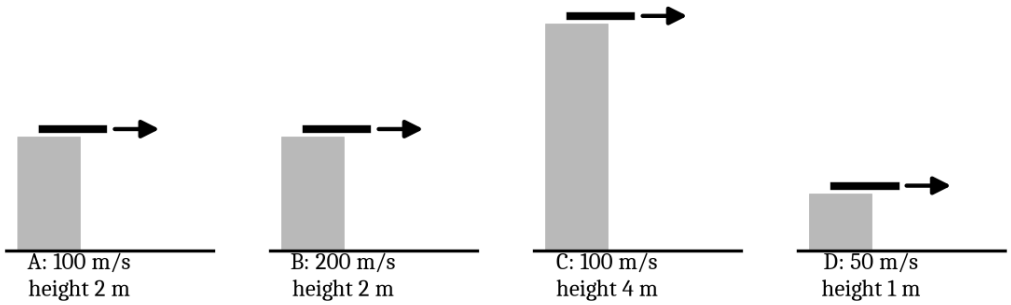
3. ○ A plane flying horizontally at 80 m/s and 500 m above the ground releases a package. How far ahead of the target must it release the package?

4. ● A ball leaves a 1.25 m table horizontally and lands 2.0 m from the table's edge. Find its launch speed.

5. ○ Mr. Jantz throws a ball horizontally off the school roof at 12 m/s. It lands 30 m from the base of the building. How tall is the building?

6. ○ **Rifle shots (TIPERs style).** Each rifle fires a bullet horizontally. Rank the times for the bullets to hit the ground, greatest first. Explain.

Ranking: _____



7. ○ If you double the launch speed of a ball rolling off a table, what happens to the time in the air? To the landing distance? If you double the table height?

8. ● Moving-cart launch. A cart rolls along a level track at a constant 1.8 m/s. A spring launcher on the cart fires a ball straight up at 5.5 m/s relative to the cart.

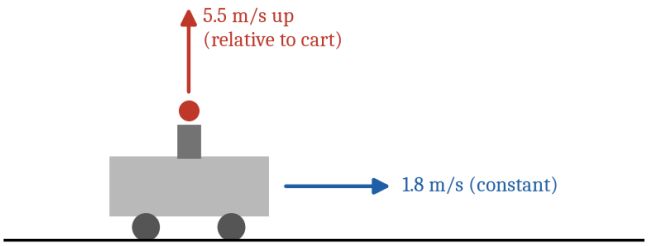
a) Describe the ball's path as seen by a rider on the cart. Where does the ball land?

b) Describe the path as seen by someone standing beside the track. What is the ball's horizontal velocity during the flight, and why?

c) Use x and y columns to find the time the ball is in the air and its maximum height above the launcher.

d) How far does the cart travel while the ball is in the air? How far does the ball travel horizontally? Where does the ball land relative to the launcher?

e) In the ground frame, what are the magnitude and direction of the ball's launch velocity?



9. ● Paragraph argument. A classmate says: “The ball in problem 8 lands behind the cart, because the cart keeps moving forward while the ball is in the air.” Do you agree? Write a paragraph argument: a claim, evidence (a physics principle or equation), and reasoning that links them, in 3–5 complete sentences.

Then answer in one sentence: where would the ball land if the cart sped up while the ball was in the air?

10. ● **Reason, then derive.** A plane flying level at speed v and height H drops a package.

a) Without equations: if the plane flies twice as high at the same speed, must the pilot release the package twice as far before the target, more than twice as far, or less? Explain in words.

b) Derive how far before the target the pilot must release the package, in terms of v, H, and g.

c) Is your expression from b) consistent with your reasoning in a)? If not, which reasoning went wrong?

d) Show that the line of sight from the plane down to the target at release makes an angle ϕ below horizontal with $\tan\phi = \dfrac{\sqrt{\frac{gH}{2}}}{v}$.

11. ○ **Qualitative.** A ball rolls off a table on the Moon ($g = 1.6\ \text{m/s}^2$) at the same speed and from the same height as on Earth. By what factor do the fall time and the landing distance change? Explain without plugging in numbers for h or v.

12. ○ A ball rolls off a 1.8 m shelf at 3.0 m/s. Find the time in the air and the landing distance.

13. ○ A stone is thrown horizontally at 8.0 m/s from a 30 m bridge. How long is it in the air, and how far from the point below the bridge does it hit the water?

14. ★ **Challenge.** Marble A rolls off a 0.80 m table at 3.0 m/s. Marble B rolls off a different table at 2.0 m/s and lands the same distance from its table. How tall is B's table?

15. ★ **Challenge.** Mr. Lehman rides a motorcycle off the end of a ramp, moving horizontally 5.0 m above the ground. The landing platform is 2.0 m tall and its near edge is 12 m away horizontally. What is the minimum launch speed?

Section 2.9 · Velocity During Flight (Marble Off the Table)

Tier: Core Skills **In class:** Marble off the table; PhET Projectile Motion

Learning targets

- C4 I can find the horizontal and vertical velocity components of a projectile at any time.

Key ideas

- At any time: $v_x = v_{ix}$ (it never changes) and $v_y = a_y t + v_{iy}$.
- Speed $v = \sqrt{v_x^2 + v_y^2}$. Direction $\theta = \tan^{-1}\left(\dfrac{|v_y|}{v_x}\right)$ below the horizontal while falling.
- Use $v_y^2 = v_{iy}^2 + 2a_y(y_f - y_i)$ when you know the drop but not the time.

1. ● A ball rolls off a 1.0 m table at 3.0 m/s.

a) Find v_x and v_y 0.20 s after it leaves the table.

b) Find the speed and direction of the ball at that moment.

c) Find the speed and direction just before it hits the floor.

2. ○ Mr. Lehman runs horizontally off a 20 m cliff at 2.5 m/s and dives into the water. Find how far out from the cliff Mr. Lehman lands and the velocity (magnitude and direction) entering the water.

3. ● **Lab prediction.** Your marble rolls 0.60 m along the tabletop in 0.40 s, then leaves the 0.85 m table. Predict where to place the cup on the floor. Write your measured landing spot after testing.

Predicted: _____ Measured: _____ Percent difference: _____

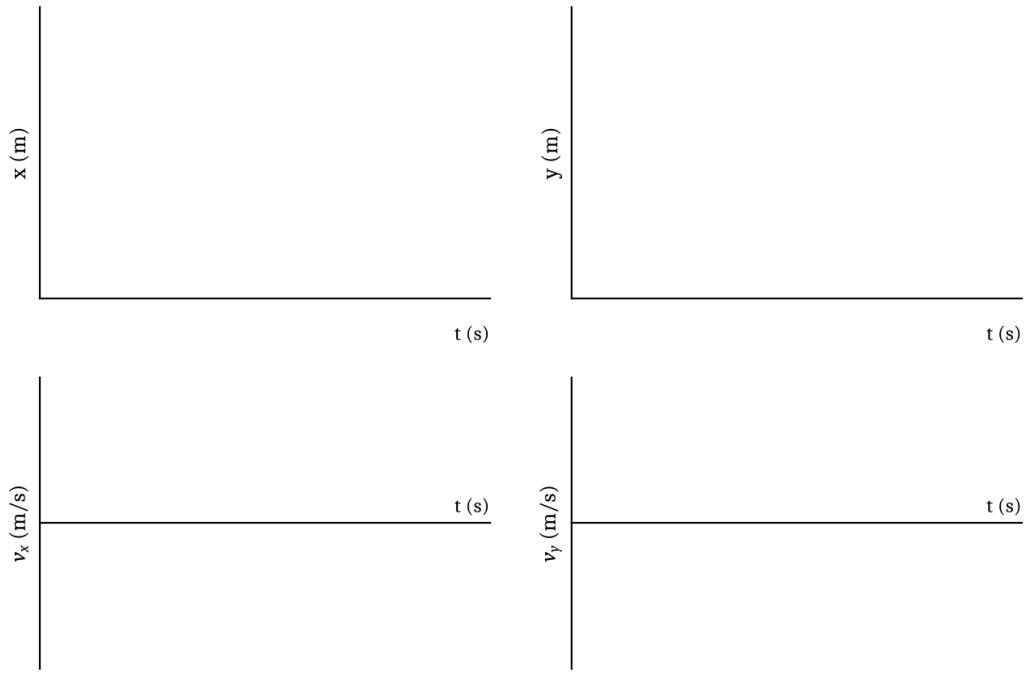
4. ○ A ball is launched horizontally at 6.0 m/s. At what time is its velocity directed 45° below the horizontal?

5. ○ Two balls roll off the same table, one fast and one slow. Which hits the floor at a steeper angle? Explain.

6. ○ **PhET check.** In the PhET Projectile Motion sim, change only the mass. What happens to the path with air resistance off? Why?

7. ● **Motion graphs.** Sketch all four graphs for the marble in problem 3 (1.5 m/s off the 0.85 m table), from the moment it leaves the table until it lands. Put $x_i = 0$ at the table edge and $y = 0$ on the floor, with up positive. Label the starting value on each graph and mark the landing time t_L.

Then: what is the slope of the x–t graph? Of the v_y–t graph? Why is the y–t graph curved?



8. ○ Halfway through its time in the air, has a horizontally launched ball fallen half the height? Explain.

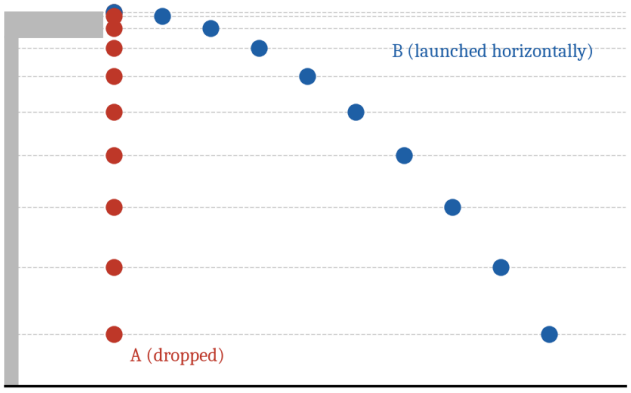
9. ○ **Symbolic.** A ball is launched horizontally at speed v from height h. Show that it hits the ground at speed $v_f = \sqrt{v^2 + 2gh}$, at an angle below horizontal with $\tan\theta = \dfrac{\sqrt{2gh}}{v}$.

10. ○ **Reading a strobe diagram.** This is the Section 2.7 picture, drawn to scale. Ball B's dots are 0.10 s apart and 0.60 m apart horizontally. B's vertical drops in the first four intervals are 0.049 m, 0.15 m, 0.25 m, 0.34 m.

a) Find B's horizontal velocity. Which of your four graphs in problem 7 shows the equal horizontal spacing?

b) Find how much each vertical drop grows from one interval to the next. What does a constant increase tell you?

c) For constant acceleration, each drop grows by $g(\Delta t)^2$. Use your answer to b) to find g.



11. ○ **Ranking.** These are the same four balls as the Section 2.7 ranking. Rank them by impact speed, greatest first.

Ball	Table height h	Launch speed v
A	1.0 m	2.0 m/s
B	1.0 m	4.0 m/s
C	4.0 m	1.0 m/s
D	0.25 m	4.0 m/s

Impact speed: Greatest _____ _____ _____ _____ Least ☐ All equal

Which matters more here, the launch speed or the table height? Explain.

12. ● Linearizing data (class lab graph). Five groups rolled marbles off the 0.85 m lab table at different launch speeds and measured the landing distance x_f from the point below the table edge.

a) Plot x_f (vertical axis) against launch speed v_{ix} (horizontal axis) on the grid. Draw one straight best-fit line.

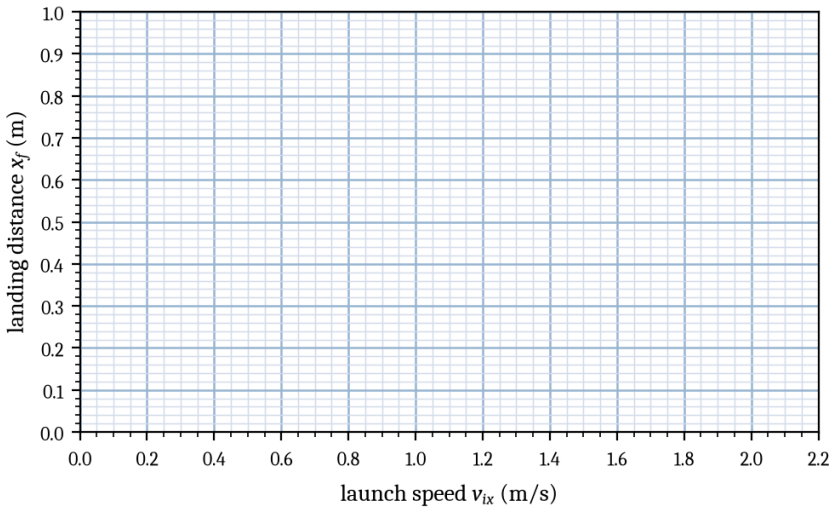
b) Find the slope of your line, using two points on the line (not data points). Include units.

c) Use the x column, $x_f = v_{ix}t + x_i$ with $x_i = 0$, to explain why the slope is the fall time t.

d) Use the calculated time from the slope of your x_f vs. v_{ix} graph to calculate g and find the percent difference from 9.8 m/s^2.

e) Why should the best-fit line pass through the origin?

Trial	**1**	**2**	**3**	**4**	**5**
Launch speed v_{ix} (m/s)	0.82	1.09	1.41	1.68	1.97
Landing distance x_f (m)	0.35	0.44	0.59	0.69	0.83



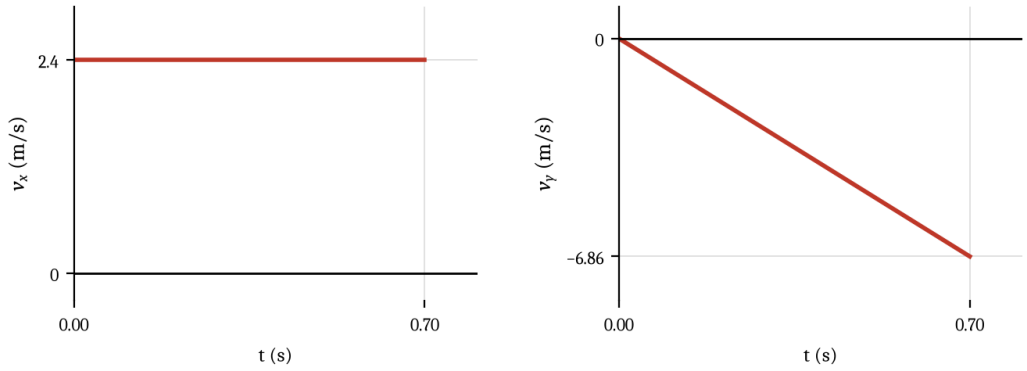
13. ● Area under a velocity–time graph. The graphs show a ball that rolls off a ledge and lands on the ground below (up positive). The area between a velocity–time graph and the time axis is the change in position; the slope is the acceleration.

a) Use the area under the v_x–t graph to find how far from the ledge the ball lands.

b) Use the area under the v_y–t graph to find the height of the ledge.

c) Use the slope of the v_y–t graph to find g.

d) Why is the area under the v_y–t graph negative? What does the sign tell you?



14. ● Design an experiment. A classmate claims that a heavier ball rolling off a table lands closer to the table because "heavy things fall faster." You have a ramp, a lab table, solid balls of the same size made of steel, glass, and wood, a balance, a meter stick, tape, and a phone with slow-motion video.

a) List the quantities you would measure and the equipment you would use for each.

b) What would you change, what would you measure, and what would you keep the same? How do you make sure every ball leaves the table at the same speed? How many trials would you run?

c) Sketch the graph of landing distance x_f vs. mass that you would expect if the claim is right, and the graph you would expect if it is wrong.

d) Your data show the steel ball landing 1 cm farther than the wood ball. Does that support the claim? Explain.

Get Ready for Quiz 2

Tier: Quiz 2: Horizontal Launches **In class:** Quiz 2: Horizontal launches (full period)

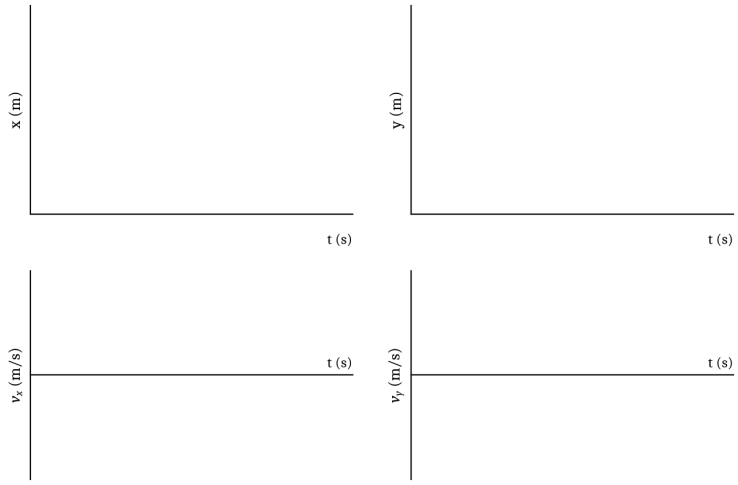
About this page

- Quiz 2 covers F4, F5, C2, C3, and C4. Complete these pages before the quiz, and rate yourself in the tracker first.

1. ● A dart is fired horizontally at 18 m/s from 1.6 m above the floor. Find the time in the air and the horizontal distance it travels.

2. ● Find the dart's velocity (magnitude and direction) just before it hits the floor.

3. ● A ball rolls off a 1.2 m table at 2.0 m/s. Sketch x vs. t, y vs. t, v_x vs. t, and v_y vs. t from the edge to the floor. Label the landing time and the starting and ending values on each graph.



4. ● True or false, and fix the false ones:

a) A projectile's horizontal velocity decreases as it falls.

b) At the moment a ball leaves the table, its acceleration is zero.

c) A heavier ball rolled off a table at the same speed lands at the same spot.

5. ● A ball rolls off a 0.95 m table and lands 1.1 m from the edge. Find its launch speed.

6. ★ **Challenge.** A stone thrown horizontally from a 20 m cliff hits the water at 25 m/s. Find its launch speed.

7. ○ **Ranking.** Four balls roll off tables. Rank them by a) time in the air and b) landing distance from the table.

Ball	Table height	Speed off the edge
A	1.0 m	2.0 m/s
B	1.0 m	4.0 m/s
C	2.0 m	2.0 m/s
D	0.50 m	4.0 m/s

a) Time in the air: Greatest _____ _____ _____ _____ Least ☐ All equal

b) Landing distance: Greatest _____ _____ _____ _____ Least ☐ All equal

Which quantity decides each ranking?

8. ○ Mr. Jantz tosses a ball horizontally at 6.0 m/s from a 15 m school roof. Write y as a function of t and graph it in Desmos. When is the ball 5.0 m above the ground, and how far from the wall is it then?

9. ○ **Symbolic.** A ball leaves a ledge of height h moving horizontally at v_i. Show that it hits the ground with speed $v = \sqrt{v_i^2 + 2gh}$. Why does the answer not depend on the ball's mass?

10. ○ A plane flying level at 60 m/s and 500 m high drops a package. Where is the plane when the package lands? Sketch the package's path as seen by someone on the ground and as seen by the pilot.

Section 2.10 · Angled Launches on Level Ground

Tier: Target **In class:** Notes and worked examples

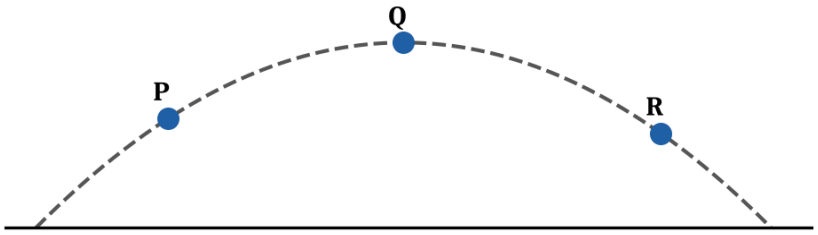
Learning targets

- T1 I can find the time of flight, maximum vertical final position, and range for an angled launch on level ground.

Key ideas

- Start with components: $v_{ix} = v_i \cos\theta$ and $v_{iy} = v_i \sin\theta$.
- At the top, $v_y = 0$ (v_x is still v_{ix}).
- $a_y = -9.8\ \text{m/s}^2$ the whole flight, including at the top.
- **Problem-solving method:** graph the y-world equation in Desmos and click the key points (see Using Desmos in the reference sheet).

1. ● On the trajectory, draw and label the velocity vector and the acceleration vector at P, Q (the top), and R.



2. ● Mr. White kicks a soccer ball at 20 m/s at 35° above level ground. Find:

a) v_{ix} and v_{iy} b) the time of flight c) the maximum height d) the range

3. ○ At the zoo, Mr. Peck films a kangaroo whose jump reaches a maximum height of 2.5 m. If it jumps at 45°, what is its launch speed and how far does it travel horizontally?

4. ○ **Launch table.** Complete the table for each launch on level ground. Show your work below the table.

Launch	v_{ix} (m/s)	v_{iy} (m/s)	Time of flight (s)	Max height (m)	Range (m)
Golf ball: 45 m/s at 40°					
Ball: 25 m/s at 30°					
Ball: 25 m/s at 60°					

Compare the 30° and 60° launches. What do you notice? Why?

5. ● **Reason, then derive.** Two balls are launched at the same speed on level ground, one at 30° and one at 60°.

a) Without equations: one student says the 60° ball goes farther because it stays in the air longer. Another says the 30° ball goes farther because it moves faster horizontally. Explain what each student gets right, and predict how the two ranges compare.

b) For speed v_i and angle θ, show that the time in the air is $T = \dfrac{2v_i \sin\theta}{g}$, the maximum height is $H = \dfrac{v_i^2 \sin^2\theta}{2g}$, and the range is $R = \dfrac{v_i^2 \sin 2\theta}{g}$ (use $2\sin\theta\cos\theta = \sin 2\theta$).

c) Is your expression for R consistent with your prediction in a)? Explain.

6. ○ **Ranking.** Four balls are launched on level ground. Fill in v_{iy} for each first, then rank them, greatest first.

Ball	Launch speed	Angle	v_{iy}
A	20 m/s	30°	
B	20 m/s	60°	
C	10 m/s	90° (straight up)	
D	14.1 m/s	45°	

a) Time in the air: Greatest _____ _____ _____ _____ Least ☐ All equal

b) Maximum height: Greatest _____ _____ _____ _____ Least ☐ All equal

c) Range: Greatest _____ _____ _____ _____ Least ☐ All equal

Explain which launch value sets each ranking.

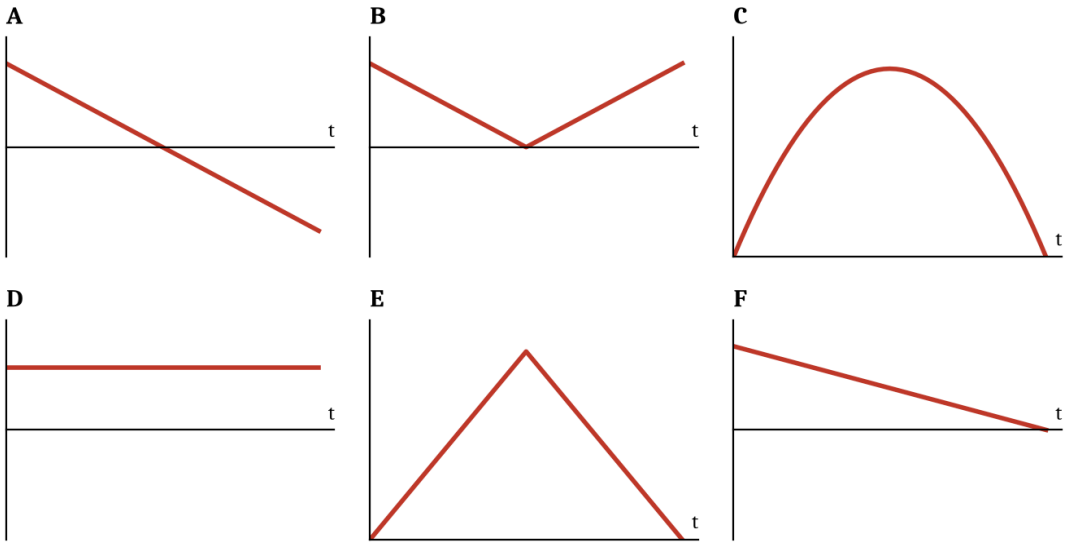
7. ● **Graphs for an angled launch.** The graphs below describe Mr. White's soccer ball in problem 2 from kick to landing (level ground, up positive). Some of them are wrong for this ball.

a) Which graph shows v_y vs. t? Which shows y vs. t?

b) Using your answers to problem 2, label on those two graphs: the starting value, the time at the top, the landing time, and the maximum height.

c) What is the slope of the v_y–t graph? What is the slope of the y–t graph at the top, and what does it tell you?

d) Which graph shows v_x vs. t? Explain its shape and give its value.



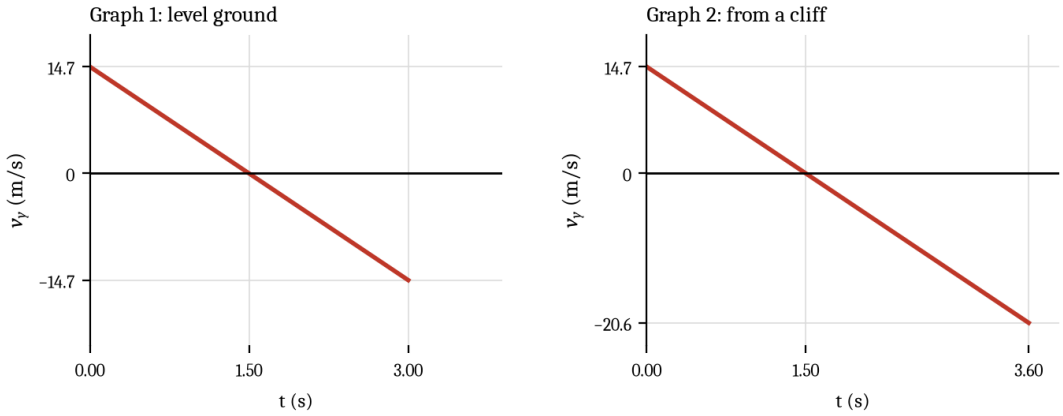
8. ● Area under a velocity–time graph (angled launch). Graph 1 is v_y vs. t for a ball launched at an angle with $v_{iy} = 14.7$ m/s that lands at its launch height. Graph 2 is the same launch from the top of a cliff; the ball lands at the base.

a) On graph 1, shade the area that equals the maximum height above the launch point, and compute it.

b) Find the total area on graph 1 (area above the axis minus area below). Explain why it is zero on level ground.

c) On graph 2, find the area above the axis and the area below the axis. Use the net area to find the cliff height.

d) Check your cliff height with the y column: $y_f = \frac{1}{2}a_y t^2 + v_{iy}t + y_i$.



9. ● Check yourself. What are the velocity and acceleration of the soccer ball in problem 2 at its highest point?

Section 2.11 · Cliffs, Ledges, and Velocity at Impact

Tier: Target **In class:** Angry Birds style problem set

Learning targets

- T2 I can solve angled launches where the final vertical position differs from the initial position.
- T3 I can find a projectile's velocity (magnitude and direction) at impact or any time.

Key ideas

- The landing time is no longer twice the time up. Use $y_f = \frac{1}{2}a_y t^2 + v_{iy}t + y_i$ and solve the quadratic.
- Launched from a height h: $y_i = h$ and $y_f = 0$ at the ground. Landing on a ledge: y_f is the ledge height, usually the later root.
- Velocity at impact: $v_x = v_{ix}, v_y = a_y t + v_{iy}$, then combine.
- **Problem-solving method:** instead of the quadratic formula, graph the y-world equation in Desmos and click where it meets the landing height.

1. ● A cannon on a 25 m cliff fires at 40 m/s, 30° above the horizontal, toward a boat at sea level.

a) Find the maximum height above the water. b) Find the time to reach the water.

c) How far from the base of the cliff is the boat? d) Find the velocity at impact.



2. ● **Angry Birds.** A bird is launched from ground level at 18 m/s at 50°. A 6.0 m tall wall stands 25 m away. How high is the bird when it reaches the wall? Does it clear the wall?

3. ○ Mrs. Luksic throws a softball at 15 m/s at 60°. It lands on a roof 6.0 m above the launch point while coming down. Find the time in the air, the horizontal distance, and the velocity at landing.

4. ● A stone is thrown from a 30 m bridge at 12 m/s, 20° **below** the horizontal. Find when and where it hits the water and its speed at impact.

5. ● **Check yourself.** For the cannon in problem 1, explain why the time to reach the water is more than twice the time to reach the top.

6. ★ **Symbolic.** A ball is launched from height h at speed v_i and angle θ and lands on the ground ($y_i = h, y_f = 0$). Use the quadratic formula to show that $t = \dfrac{v_i \sin\theta + \sqrt{v_i^2 \sin^2\theta + 2gh}}{g}$. Check that $h = 0$ gives the level-ground answer from Section 2.10.

7. ● **Monkey and hunter (concept).** A dart gun is aimed straight at a target hanging from a branch. The target is released at the instant the dart is fired. Ignoring air resistance, does the dart pass above the target, hit it, or pass below it? Explain by comparing each object with where it would be without gravity.

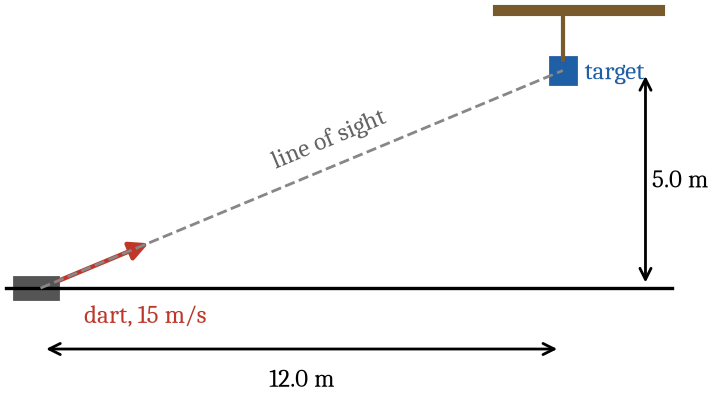
8. ○ **Monkey and hunter (numbers).** The target hangs 12.0 m away horizontally and 5.0 m above the dart's launch height. The dart is aimed straight at it and leaves at 15 m/s; the target drops at the same instant. Use x and y columns.

a) Find the launch angle, v_{ix}, and v_{iy}. b) How long does the dart take to reach the target's x position?

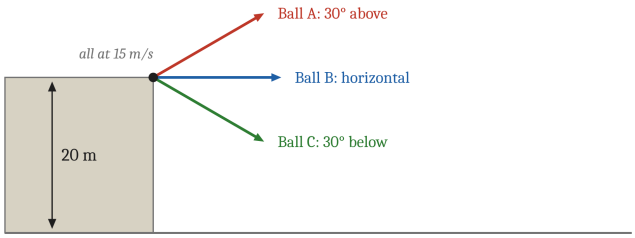
c) At that time, how far below the line of sight is the dart? How far has the target fallen?

d) Where do they meet (height above the launch point)?

e) What is the slowest launch speed that still hits the target before it falls to the launch height?



9. ○ **Ranking.** Dr. Shafer throws three medicine balls from the top of a 20 m cliff, each at 15 m/s, in the directions shown. Rank them, greatest first.



a) Time in the air: Greatest _____ _____ _____ Least ☐ All equal

b) Speed at impact: Greatest _____ _____ _____ Least ☐ All equal

Check one impact speed with components. Does the result surprise you? Explain.

Section 2.12 · Working Backward and Deriving in Symbols

Tier: Target **In class:** Marble ramp "land in the cup" challenge

Learning targets

- T4 I can work backward to find the launch speed or angle needed to hit a target, including in symbols.

Key ideas

- Write what you know and what you want in each column. Find which column gives you t, or link the columns with t.
- Deriving in symbols: keep v_i, θ, g, and heights as letters until the very end. Check units and limiting cases.
- Write g as a positive number (9.8 m/s^2) and put the minus sign in the equation: $a_y = -g$.
- **Problem-solving method:** check a numerical answer by graphing the y-world equation in Desmos.

1. ● **Derive.** A ball is launched at speed v_i and angle θ on level ground. Show that the time of flight is $t = \dfrac{2v_i \sin\theta}{g}$ and the maximum height is $y_{\max} = \dfrac{v_i^2 \sin^2\theta}{2g}$.

2. ● **Derive.** A ball thrown horizontally at speed v_i from the top of a cliff lands a horizontal distance d from the base. Find the cliff height h in terms of v_i, d, and g.

3. ● **Land in the cup.** A marble leaves the end of a horizontal ramp 0.95 m above the floor and lands 0.80 m out. What was its launch speed?

4. ○ Your ramp launches marbles horizontally at 3.2 m/s from 1.10 m above the floor. The cup is 0.10 m tall. Where should you place the cup?

5. ○ A ball launched at 20 m/s must land on level ground 30 m away. Find both launch angles that work.

6. ★ **Free throw challenge.** Mr. Vincenzo's free throw leaves his hands 2.0 m above the floor at 50°. The rim is 3.05 m high and 6.0 m away horizontally. What launch speed sinks the shot?

7. ● **Limiting cases.** Check the expressions you derived in problems 1 and 2.

a) What do $t = \dfrac{2v_i \sin\theta}{g}$ and $y_{\max} = \dfrac{v_i^2 \sin^2\theta}{2g}$ predict as $\theta \to 0$? Does that make physical sense?

b) What do they predict at $\theta = 90^\circ$? Compare with a ball thrown straight up at v_i. What is the range then?

c) In $h = \dfrac{gd^2}{2v_i^2}$, what happens to h as v_i becomes very large while d stays fixed? Does that make sense?

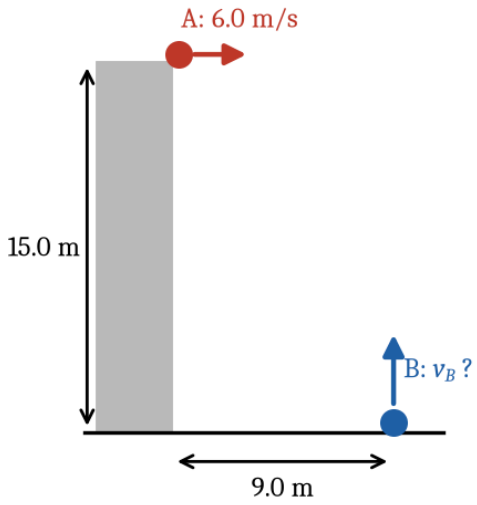
8. ● Two projectiles meet. Ball A is thrown horizontally at 6.0 m/s from the top of a 15.0 m tower. At the same instant, ball B is thrown straight up from the ground 9.0 m from the base of the tower.

a) Use A's x column to find when A is directly above B's launch spot.

b) Set the two vertical positions equal at that time. Show that the $\frac{1}{2}a_y t^2$ terms cancel, and find B's launch speed so the balls collide.

c) How high above the ground do they meet? Is B moving up or down then? Has A already landed?

d) Show in symbols that $v_B = \dfrac{H v_A}{d}$.



9. ● Paragraph argument. A classmate claims that on the Moon ($g = 1.6\ \text{m/s}^2$), ball B in problem 8 would need a different launch speed to hit ball A. Do you agree? Write a paragraph argument: a claim, evidence (a physics principle or equation), and reasoning that links them, in 3–5 complete sentences.

Get Ready for Quiz 3

Tier: Quiz 3: Angled Launches **In class:** Quiz 3: Angled launches (full period)

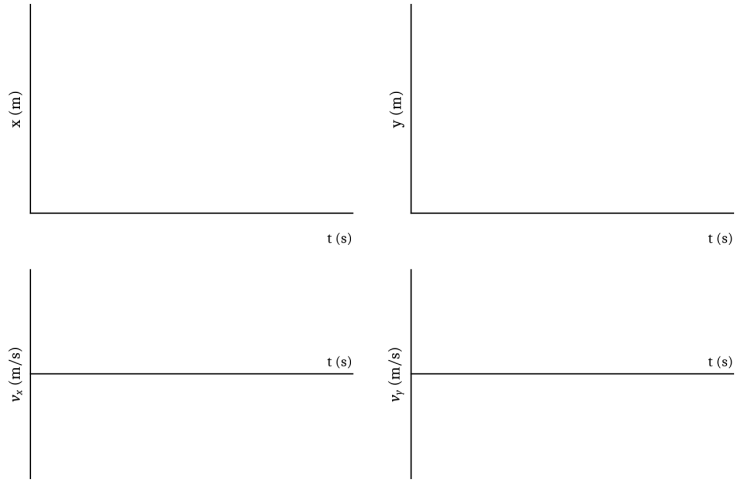
About this page

- Quiz 3 covers T1 through T4. Complete these pages before the quiz, and rate yourself in the tracker first.

1. ● Mr. White punts a football at 24 m/s at 55° on level ground. Find the hang time, maximum height, and range.

2. ● Dr. Shafer heaves a medicine ball from a 12 m roof at 15 m/s, 30° above the horizontal. Find the time in the air, the horizontal distance, and the velocity at impact.

3. ● A ball is launched at 20 m/s, 30° above level ground. Sketch x vs. t, y vs. t, v_x vs. t, and v_y vs. t for the whole flight. Label v_{ix}, v_{iy}, the time at the top, and the landing time.



4. ★ **Challenge.** A ball is launched at 20 m/s on level ground and lands 30 m away. Find both launch angles that work.

5. ● Mr. White kicks a ball at 18 m/s at 35° on level ground. Find v_x and v_y at the top and the time to get there.

6. ○ Mrs. Luksic throws a softball from 1.5 m above the ground at 20 m/s, 40° above the horizontal, toward a building with a flat roof 8.0 m high. The front edge of the roof is 25 m away horizontally. Does the ball clear the edge? If it lands on the roof, when and where? Your y equation has two solutions for $y_f = 8.0$ m. What does the other one mean?

7. ○ Mr. Vincenzo shoots a basketball from 2.1 m above the floor at 7.5 m/s, 52° above the horizontal. The rim is 3.05 m high and 4.2 m away horizontally. How high is the ball when it is directly above the rim? Should Mr. Vincenzo shoot a little harder or a little softer? Explain.

8. ● **Ranking.** Four balls are launched at 20 m/s on level ground at the angles below. Rank them by a) maximum height and b) range.

Ball	Launch angle
A	15°
B	30°
C	60°
D	75°

a) Maximum height: Greatest _____ _____ _____ _____ Least ☐ All equal

b) Range: Greatest _____ _____ _____ _____ Least ☐ All equal

Explain why two pairs tie in b).

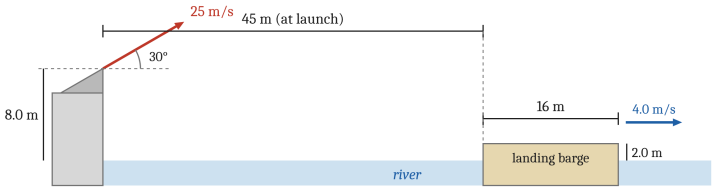
9. ○ Mr. White puts a shot at 13 m/s from 2.1 m above the ground. Find the horizontal distance at 40° and at 45°. Which angle is better, and why isn't 45° automatically best here?

10. ○ **Symbolic.** A ball is launched at speed v_i and angle θ. Write expressions for v_x and v_y at any time t, and for the ball's speed at the top of its path. Why is the speed at the top not zero?

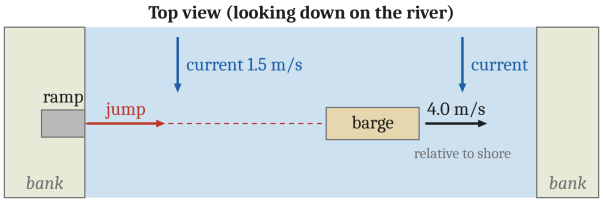
Putting It All Together

Tier: Target **In class:** Two full-page challenge problems; read the whole story and sketch it first

1. ● Mr. Lehman's River Jump. Mr. Lehman is jumping his motorcycle across a river for a charity show. The ramp edge is 8.0 m above the water and launches him at 25 m/s, 30° above the horizontal. He must land on a barge whose deck is 2.0 m above the water and 16 m long. At the instant he leaves the ramp, the near edge of the deck is 45 m from the ramp edge, and the barge is moving straight away from the ramp at a steady 4.0 m/s relative to the shore. Use down as negative and put the water at $y = 0$.



a) **Steering the barge.** The river's current flows at 1.5 m/s along the banks, sideways to the jump (see the top view). To move straight away from the ramp at 4.0 m/s relative to the shore, which way must the barge point, and how fast must it move through the water? Sketch the vectors.

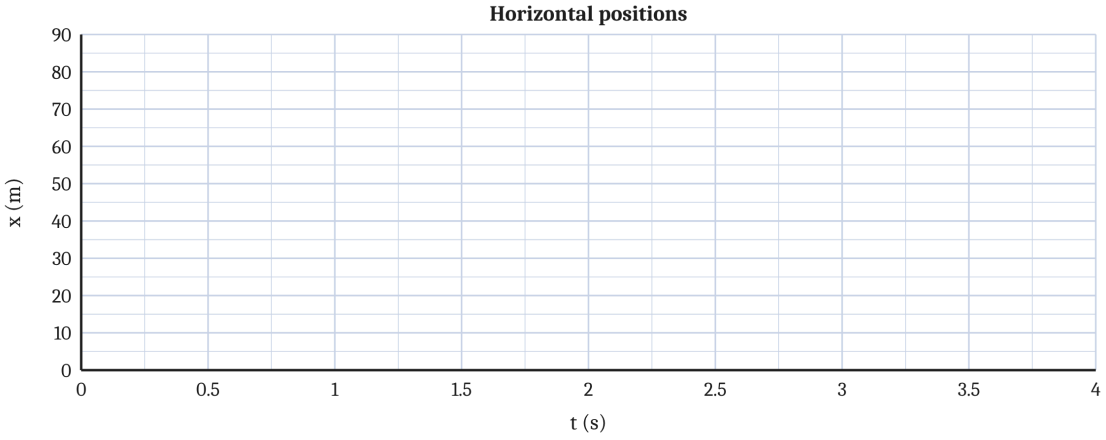


b) Does Mr. Lehman land on the deck? Support your answer with calculations, and say where on the deck he touches down.

c) Would the stunt have worked if the barge had stayed still? Explain.

d) How fast is Mr. Lehman moving, and in what direction, relative to the barge as he touches down?

Graph the jump. Put the ramp edge at $x = 0$. Use 4.0 m/s for the barge on every graph.



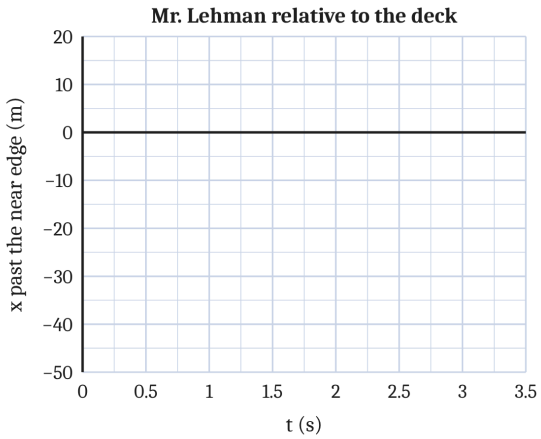
e) Graph Mr. Lehman's horizontal position and the positions of the deck's near edge and far edge from $t = 0$ to $t = 4.0$ s. Draw his line solid until touchdown, then dashed, as if he kept flying. Circle his touchdown point.

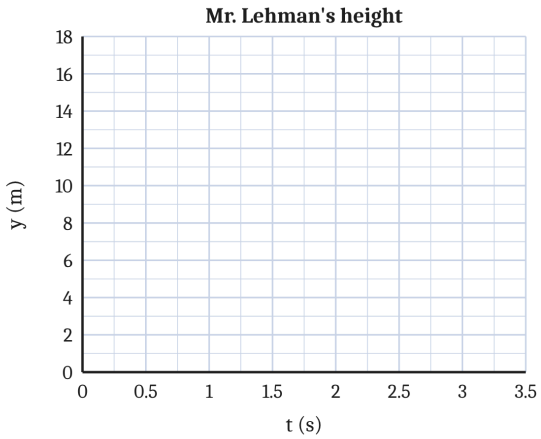
f) Find where Mr. Lehman's line crosses each barge line, first from your graph, then by setting his position equation equal to each edge's position equation. What is happening at each crossing?

g) Use the two crossing times and his touchdown time to explain why he lands on the deck. What would a touchdown before the first crossing mean? After the second?

h) What does the slope of each line stand for? Why are the two barge lines parallel, and what would they look like if the barge stayed still?

The jump as seen from the barge.



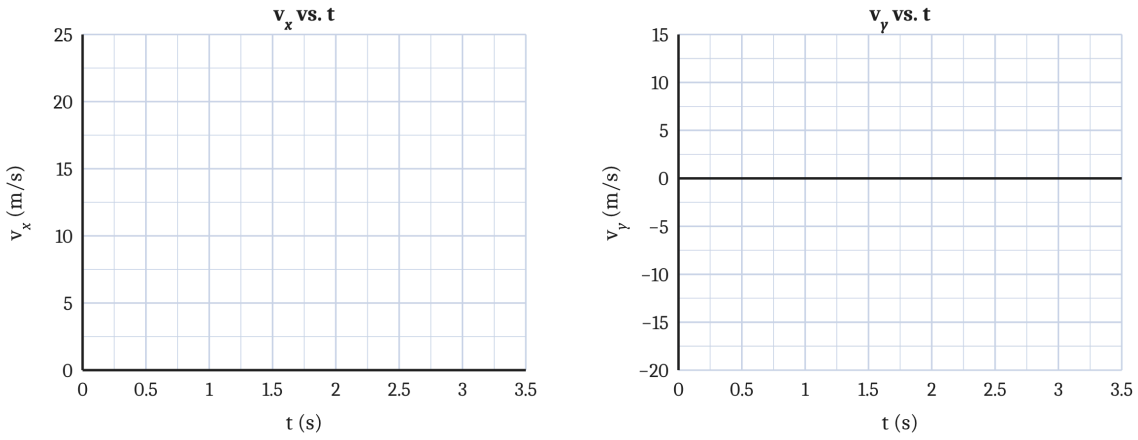


i) On the left grid, graph Mr. Lehman's horizontal position measured from the deck's near edge (positive past the edge), from launch until touchdown. Where does your line cross zero, and how does that match part f? What range of values at touchdown means he lands on the deck?

j) Why is this line straight, and what does its slope stand for? Compare it with your answer to part d.

k) On the right grid, graph Mr. Lehman's height above the water and draw the deck height as a dashed line. Mark the top of the jump and the touchdown. How does the graph show his time in the air?

Velocity graphs.



l) Graph Mr. Lehman's v_x and v_y from launch until touchdown.

m) Use the area between each line and the time axis to find Δx and Δy for the jump. The v_y graph has area above and below the axis: what does each part stand for, and why does the total match the drop from the ramp edge to the deck?

n) The producer wants him to touch down in the middle of the deck. Recommend one change to the setup, calculate the new value, and describe how your horizontal-position graph would change.

o) **Challenge.** What range of barge speeds lets Mr. Lehman land on the deck? Show the fastest and slowest cases as extra lines on your horizontal-position graph.

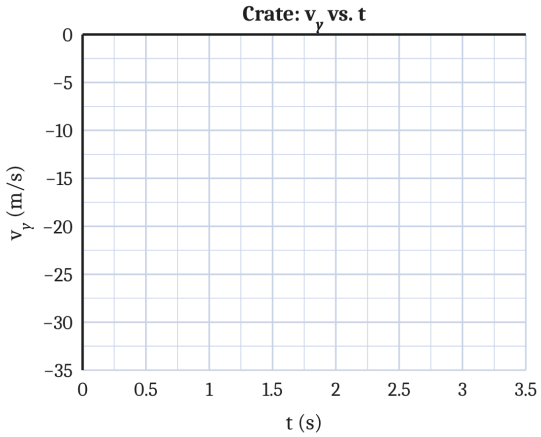
2. ● Mountain Rescue. Dr. Shafer tried to climb a mountain and got stuck. Mr. Jantz is flying a rescue helicopter to Dr. Shafer's camp, 30 km due north of the base. The helicopter's airspeed is 50 m/s, and a 12 m/s wind blows from west to east the whole time.

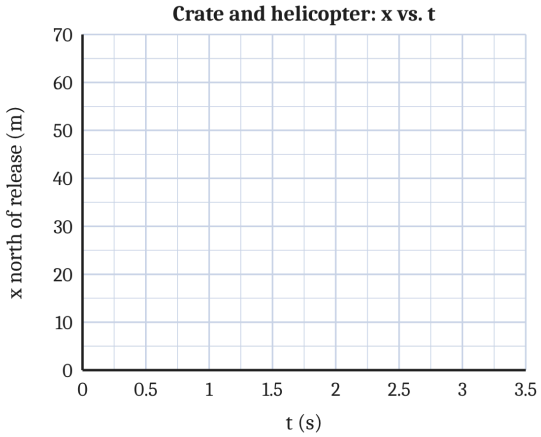
a) Which way must Mr. Jantz point the helicopter to travel due north, and how many minutes does the trip take? Sketch the vectors.

b) Near the camp he flies due north at a ground speed of 20 m/s, 45 m above the ground, and releases a supply crate. Where must he release it so it lands at the camp, and how fast is the crate moving when it lands?

c) A gust carries the crate into the trees. Dr. Shafer reaches it by following a trail: 30 m at 20° E of N, then 25 m at 60° E of N. How far, and in what direction, should Dr. Shafer walk to get straight back to camp?

Graph the crate drop. Start the clock when the crate is released.





d) On the left grid, graph the crate's v_y from release to landing. Use the area between your line and the time axis to check the 45 m drop.

e) On the right grid, graph the horizontal position of the crate and of the helicopter (still flying north at 20 m/s) from release until the crate lands. What do your two lines show about where the crate is, compared with the helicopter, while it falls?

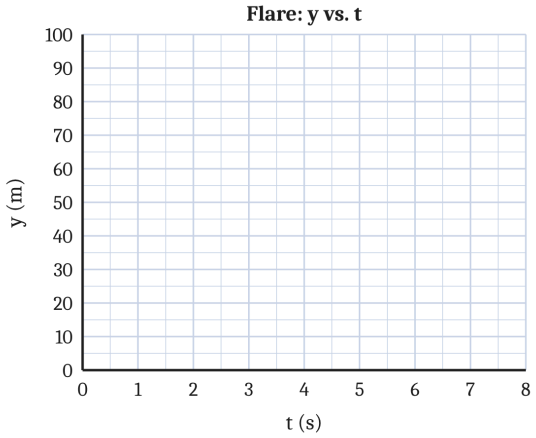
f) A real crate feels air resistance. Sketch how the crate's line on the right grid would change. Should Mr. Jantz release it earlier or later than in part b? Explain using your graph.

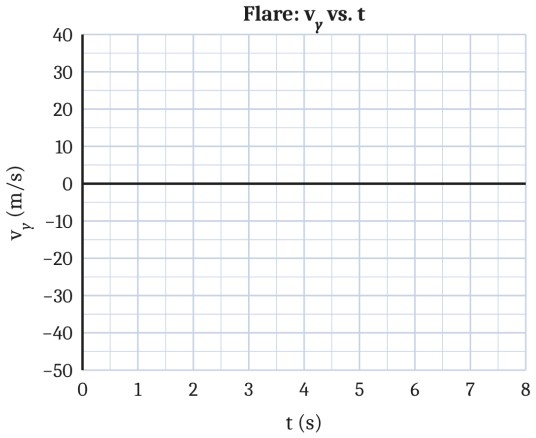
The signal flare. That night Dr. Shafer stands at the edge of a 25 m cliff and fires a flare at 40 m/s, 60° above the horizontal. Mr. Jantz hovers 70 m above the valley floor. Use the valley floor as $y = 0$.



g) Does the flare rise high enough to pass Mr. Jantz's altitude? If so, for how long is it above that altitude, and where does it land?

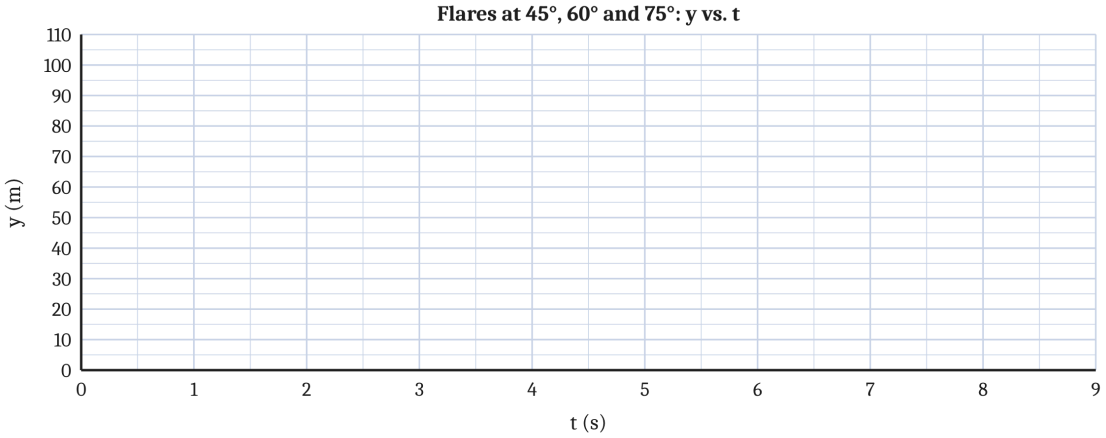
h) On the left grid, graph the flare's height from launch to landing (plot points or copy your Desmos graph). Draw a line at 70 m and mark the crossings. Draw a tangent line at each crossing and find its slope. What do the slopes stand for, and how do they compare?



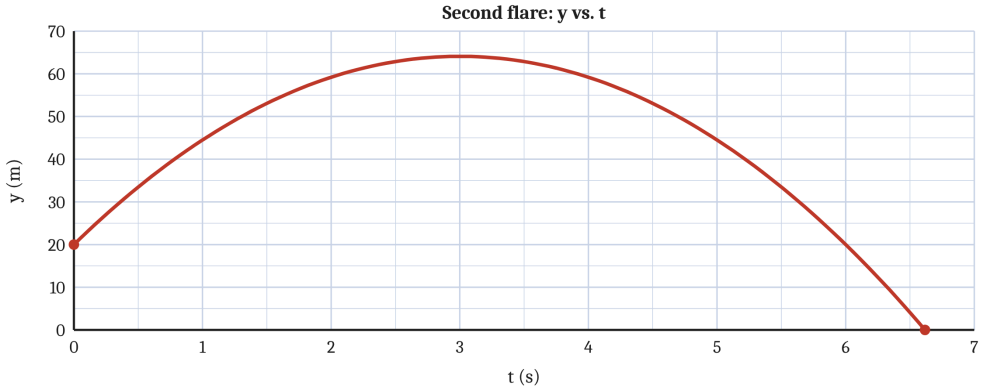


i) On the right grid, graph the flare's v_y from launch to landing. Shade the area between your line and the time axis from the first crossing of 70 m to the second. What is the total of that area, and why?

j) Dr. Shafer could fire at 45° or 75° instead. Sketch the 45°, 60° and 75° flares on the grid (find each one's v_{iy}, highest point and landing time first). Using your graph, decide which angle keeps the flare above 70 m the longest and whether every angle reaches 70 m. What feature of the graph did you use?



Reading a graph. The next night Dr. Shafer fires a second flare at 40 m/s from a different ledge. Mr. Jantz's tracker records the graph below. Use only the graph and the launch speed.



k) How high is the ledge? Find v_{iy} two ways: from the time it takes to reach the top, and from the height the flare gains.

l) At what angle above the horizontal was the flare fired?

m) About how far from the base of the ledge does the flare land?

Target tracker

After each section and before each quiz, rate yourself: ✓ secure, ~ getting there, × need help. Write the quiz or reassessment result when you get it back.

Tier	Target	I can...	Section	Self-rating	Quiz / test evidence
Foundations	F1	Draw vectors to scale and add them tip-to-tail	2.2		
Foundations	F2	Resolve a vector into x and y components (SOH-CAH-TOA)	2.3		
Foundations	F3	Find a resultant's magnitude and direction	2.2, 2.3		
Foundations	F4	Explain why horizontal and vertical motions are independent	2.7		
Foundations	F5	State v_x is constant and $a_y = -9.8\ \text{m/s}^2$	2.7		
Foundations	F6	Find the resultant of two perpendicular vectors	2.2		
Core Skills	C1	Add non-perpendicular vectors using components	2.4		
Core Skills	C2	Organize a projectile problem into x and y columns	2.8		
Core Skills	C3	Solve horizontal launches: time to land, landing position	2.8		
Core Skills	C4	Find velocity components at any time	2.9		
Core Skills	C5	Relative velocity with perpendicular vectors	2.5		
Target	T1	Angled launch on level ground: time, max height, range	2.10		
Target	T2	Angled launch from a cliff or onto a ledge	2.11		
Target	T3	Velocity at impact or any time	2.11		
Target	T4	Work backward to a launch speed or angle	2.12		
Target	T5	Relative velocity with angles	2.6		

Extension menu

Open to students who have shown the Target tier. Choose one. The Projectile Launch Project is the class-wide predict-then-test lab; these go further.

Air resistance

Film a badminton birdie or foam ball, track its path with video analysis, and compare it to the ideal model. Explain where and why the real path differs.

Range equation

Derive $R = \dfrac{v_i^2 \sin 2\theta}{g}$ for level ground. Show why 45° gives the maximum range and why complementary angles (such as 30° and 60°) give equal ranges. Build a spreadsheet model to check.

Sports ballistics

Explain why the best release angle for a shot put or a basketball shot is not 45°, using calculations with a raised initial position.

My choice: _____ Due date: _____